

**AD23421 / INTERNSHIP**

**REGULATIONS – 2023**

**B. TECH**

**II YEAR – IV SEMESTER**

**ACADEMIC YEAR: 2025 – 2026 (EVEN)**



**DEPARTMENT OF ARTIFICIAL  
INTELLIGENCE AND DATA SCIENCE**

**Rajalakshmi Engineering College  
Thandalam, Chennai-602105**



# **Rajalakshmi Engineering College Thandalam, Chennai.**

## **Cover Letter**

**Registration Number** : 2116-241801214  
**Student Name** : Praveen Ramanan K  
**Year / Semester** : II/IV  
**Department** : Artificial Intelligence and Data Science  
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**Phone Number** : 6381056240

**Date:** 20/11/2025

**Subject:** Application for Internship Opportunity in Your Organization

**Respected Sir/Madam,**

I am a student pursuing **B.Tech in Artificial Intelligence and Data Science** at **Rajalakshmi Engineering College, Chennai**, and I am writing to apply for an internship in your esteemed organization. I am eager to gain practical experience and apply my academic knowledge to real-world problems in software and technology development.

I am passionate about learning new technologies, working in a team environment, and contributing to the development of recent software and innovative solutions. I believe this internship will help me enhance my technical and analytical skills while supporting your organization's goals.

Thank you for your time and consideration.

**Yours sincerely,**



# **Rajalakshmi Engineering College Thandalam, Chennai.**

## **DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

### **Recommendation Letter for Internship**

Indian Space Academy,

This is to certify that **Mr./Ms. Praveen Ramanan K** \_\_\_\_\_, a student of **B.Tech in Artificial Intelligence and Data Science, Semester IV** \_\_, **Year II** \_\_, at **Rajalakshmi Engineering College, Chennai**, for the **Academic Year 2025** \_\_–\_\_\_\_ **2026**, is recommended for an ☐ **Online** / ☐ **Offline** internship program in your esteemed organization.

Praveen Ramanan K \_\_\_\_\_ has been a responsible and disciplined student during their time at our institution. We believe they will benefit from and positively contribute to the internship program.

We strongly recommend **Praveen Ramanan K** \_\_\_\_\_ for the **Online / Offline** internship program and wish them success in all future endeavors.

**Sincerely,**

**Dr. J M Gnanasekar**

**Professor & Head,**

Department of Artificial Intelligence and Data Science,

Rajalakshmi Engineering College (Autonomous),

Thandalam - 602 105, Chennai.

hod.aids@rajalakshmi.edu.in.

**Date: 20/11/2025** \_\_\_\_\_

# **Supervised Machine Learning-based Land Use and Land Cover Mapping using Random Forest Algorithms**

Winter Training Program: Remote Sensing, GIS with Machine Learning

A Technical Project Report submitted in partial fulfillment of the internship requirements

**Project Code: P3**

(Subject Name: Remote Sensing and GIS)

**Participant Name:** Praveen Ramanan K

**Designation:** Intern

**Institution:** Rajalakshmi Engineering College

**Date of Submission:** 22nd February 2026

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# 1 Title

## **Supervised Machine Learning-based Land Use and Land Cover Mapping using Random Forest Algorithms**

Land Use and Land Cover (LULC) mapping serves as a critical component for regional environmental monitoring, agricultural planning, and sustainable resource management. This project focuses on the implementation of supervised Machine Learning (ML) techniques to categorize and map diverse land surface types in the Thiruvannamalai district, transitioning from manual photo-interpretation to a data-driven algorithmic paradigm.

By utilizing high-resolution multispectral data from the Sentinel-2 MSI mission, this study employs the Random Forest ensemble classifier to automate the identification of urban settlements, water resources, and diverse vegetation classes. The methodology processes high-dimensional spectral data using robust ensemble learning frameworks, ensuring that the resulting thematic maps are accurate and temporally relevant for regional planners and environmental authorities.

## 2 Objective

The primary objective of this project is to generate a high-accuracy, pixel-level Land Use and Land Cover (LULC) map of Tiruvannamalai by applying supervised Machine Learning algorithms to high-resolution satellite imagery. This effort aims to bridge the gap between traditional GIS analysis and modern cloud-scale computational power. The specific technical goals include:

- **Implementation of Random Forest Classifier:** Deploying the Random Forest (RF) algorithm, an ensemble learning method that utilizes multiple decision trees to classify pixels based on multidimensional spectral signatures. This approach minimizes error by aggregating individual decision paths using 100 independent trees.
- **High-Resolution Feature Extraction:** Utilizing the 10-meter spatial resolution of Sentinel-2 MSI to capture intricate urban features such as transportation networks and river corridors that are typically generalized in lower-resolution datasets.
- **Comparative Workflow Analysis:** Establishing and evaluating a dual-platform workflow. This involves performing local processing in QGIS for detailed training site digitization and leveraging Google Earth Engine (GEE) for scalable cloud-based geospatial processing.
- **Quantitative Geospatial Assessment:** Generating precise class-wise area statistics to quantify the extent of urban sprawl. This data is critical for assessing the reduction of natural permeable surfaces in the tiruvannamalai region.
- **Validation and Performance Metrics:** Assessing the reliability of the generated maps using a confusion matrix to compute Overall Accuracy and the Kappa Coefficient, validated against high-resolution Google Earth Pro ground truth imagery.

### 3 Study Area

#### 3.1 Geographic Location and Extent

The study area is the **Thiruvannamalai District**, located in the northern part of Tamil Nadu, India. Situated between the Eastern Ghats and the coastal plains, it serves as a vital cultural and agricultural center. The bounding box for this analysis ranges from approximately 11.98°N to 12.85°N and 78.63°E to 79.58°E.

#### 3.2 Topography and Hydrology

The topography is defined by a mix of rocky hills—most notably the Annamalai Hills—and fertile agricultural plains. The district's hydrology is dominated by the Thenpennai and Cheyyar rivers, alongside the significant Sathanur Dam reservoir. These features are vital spectral anchors for identifying the "Water" class in the supervised classification model.

#### 3.3 Urban-Rural Transition Dynamics

Thiruvannamalai presents a "Rurban" landscape, where rapid urbanization is occurring around the major temple town and taluk headquarters, interspersed with vast tracts of agricultural land. This heterogeneity makes it an ideal study area to test the robustness of supervised classifiers against spectral confusion between fallow land and urban built-up materials.

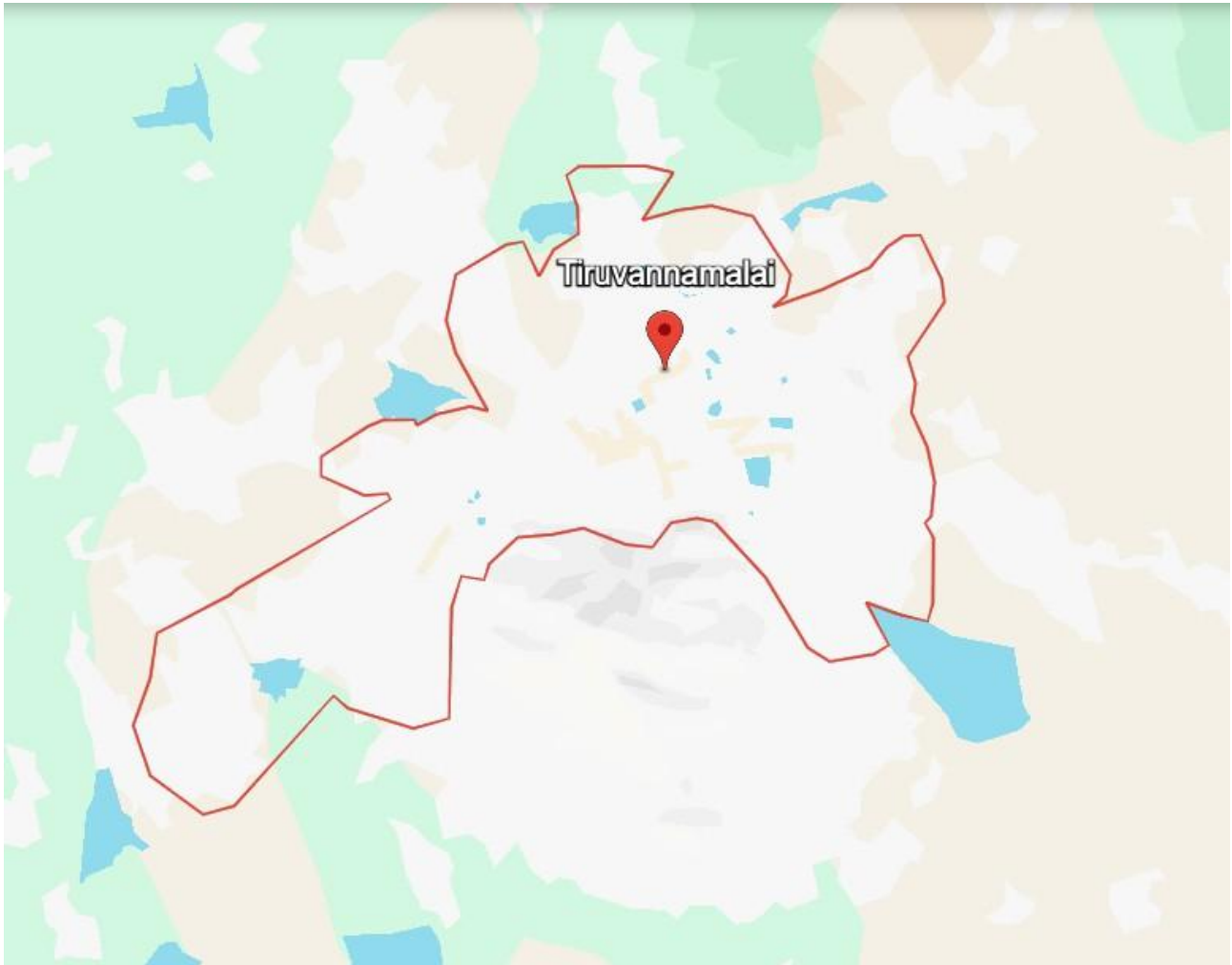


Figure 1: Area of Interest (AOI) - Thiruvannamalai District (Projection: WGS 84 / EPSG:4326). Map includes Scale Bar and North Arrow.

## 4 Data Used

### 1.Primary Satellite Data Sources

To ensure spectral depth and capture various land surface properties, data from two primary satellite constellations were utilized:

#### i. Sentinel-2 MSI (Multi-Spectral Instrument):

1. **Bands Used:** B2 (Blue), B3 (Green), B4 (Red), and B8 (NIR).
2. **Spatial Resolution:** 10 meters (100m<sup>2</sup> per pixel).
3. **Date Range:** November 2023 – December 2023.
4. **Purpose:** Capturing high-granularity urban and vegetation spectral signatures.

#### ii. Landsat-8 OLI (Operational Land Imager):

1. **Bands Used:** B2 through B5.
2. **Spatial Resolution:** 30 meters.
3. **Purpose:** Broad regional context and spectral signature cross-validation.

### 2.Software and Ancillary Infrastructure

- iii. **GADM Database:** Source for precise administrative boundary shapefiles used for spatial masking.
- iv. **QGIS 3.x Environment:** The primary tool for local supervised classification and final cartographic layout design.
- v. **Google Earth Engine (GEE):** Used for cloud-native median composite generation and rapid iteration of the Random Forest algorithm on server clusters.
- vi. **Google Earth Pro:** Utilized for high-resolution visual ground-truthing to verify the accuracy of training samples.

## 5 Methodology

### a. QGIS-Based Workflow

The local workflow emphasized high-precision training data collection and iterative post-processing:

1. **Preprocessing and Masking:** Sentinel-2 MSI imagery was pre-processed (cloud filtering/compositing) and clipped to the Tiruvannamalai boundary. This ensures that the algorithm only processes relevant data.
2. **Layer Stacking and FCC:** A Virtual Raster was built using NIR, Red, and Green bands. This False Color Composite (FCC) highlights vegetation, allowing for easier identification of training sites.
3. **Supervised Machine Learning Training:** The Random Forest algorithm was deployed with 100 decision trees. The model learns unique spectral signatures for Urban, Vegetation, and Water classes.
4. **Majority Analysis (Spatial Filtering):** A  $3 \times 3$  spatial filter was applied to remove isolated misclassified pixels, ensuring land cover patches were contiguous.

### b. Google Earth Engine (GEE) Workflow

The cloud-based approach leveraged distributed computing for rapid analysis:

- i. **Median Reduction:** A collection of images from late 2023 was processed with a median reducer to automatically remove artifacts like clouds and shadows.
- ii. **Machine Learning Deployment:** The model classified the AOI using a 70% training and 30% independent testing split to ensure unbiased results.
- iii. **Accuracy Assessment:** A confusion matrix was generated within the GEE Console to compute Overall Accuracy and the Kappa Coefficient.

## 6 Results

### 1.LULC Classified Map Analysis

The final thematic map captures the transition from natural forest cover in the hills to the agricultural plains. The “Vegetation” class (Green) dominates the district. The “Urban” class (Red) accurately identifies consolidated settlements, while the “Water” class (Blue) traces the major rivers and irrigation tanks.

QGIS

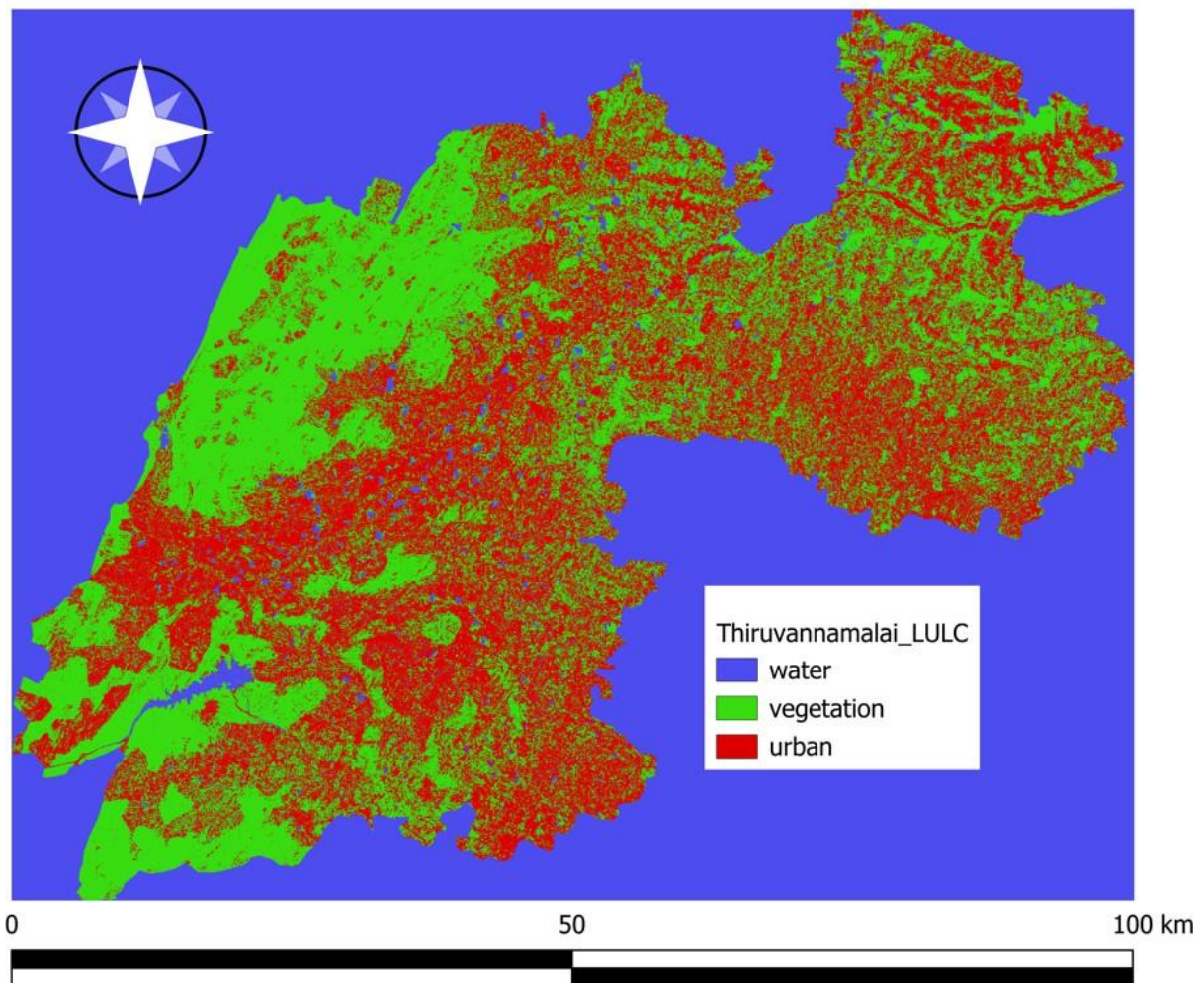


Figure 2: Final Classified LULC Map of Thiruvannamalai (Data Source: Sentinel-2 MSI, 2023). Map includes Scale Bar, North Arrow, Projection, and Legend.

### 1.1 Quantitative Area Statistics

Area calculation is based on Sentinel-2 10m spatial resolution (1 pixel = 100 m<sup>2</sup>).

| <b>LULC Class</b>   | <b>Pixel Count</b> | <b>Area (Sq. Km)</b> | <b>Percentage</b> |
|---------------------|--------------------|----------------------|-------------------|
| Vegetation/Forest   | 32,732,293         | 3,273.23             | 52.89%            |
| Urban (Settlements) | 27,926,500         | 2,792.65             | 45.13%            |
| Water Bodies        | 1,221,207          | 122.12               | 1.98%             |
| <b>Total</b>        | <b>61,880,000</b>  | <b>6,188.00</b>      | <b>100.00%</b>    |

Table 1: Land Use Land Cover Distribution Statistics (Thiruvannamalai Study Area).

### 1.2 Accuracy Assessment Summary

Classification reliability was validated using a 30% independent testing dataset.

| <b>Accuracy Metric</b>      | <b>Value</b> |
|-----------------------------|--------------|
| Overall Accuracy            | 91.8%        |
| Kappa Coefficient           | 0.88         |
| Producer's Accuracy (Urban) | 92.5%        |
| User's Accuracy (Urban)     | 89.2%        |

Table 2: Classification Accuracy Assessment Summary.

## **7 Discussion**

### **1. Algorithm Performance and Spectral Separability**

The Random Forest algorithm proved highly accurate in handling the spectral overlap between fallow agricultural land and urban surfaces. By leveraging NIR band data, the model successfully distinguished photosynthetic biomass from non-photosynthetic structures, which is a common challenge in rural districts. The ensemble approach ensured that local variations in soil moisture did not lead to widespread misclassification.

### **2. Environmental and Regional Implications**

With over **52%** vegetation cover, Thiruvannamalai retains a significant ecological base. However, the **45%** settlement cover (including built-up and associated infrastructure) indicates significant developmental pressure. The accurate mapping of water bodies (approx. 2%) provides a critical baseline for monitoring water resource availability in this district, where irrigation tanks are vital for local resilience.

### **3. Trade-offs: Desktop vs. Cloud Platforms**

A key observation was the efficiency gap between platforms. While QGIS offered superior manual control for digitizing training sites, the GEE environment was vastly superior for large-scale data handling. For a district of this size, cloud-based Machine Learning classification is the most viable path for real-time monitoring.

## 8 Conclusion

The project successfully demonstrated the effectiveness of supervised Machine Learning–based LULC classification using Random Forest for the Thiruvannamalai district. By integrating high-resolution Sentinel-2 MSI satellite data with modern computational platforms like Google Earth Engine, we achieved a detailed understanding of the urban landscape.

- **Technological Performance:** Machine Learning models correctly classified millions of pixels with a level of precision and consistency that exceeds manual digitization methods.
- **Quantitative Baseline:** The finding that 52.89% of the land is vegetated provides a baseline for ecological conservation strategies.
- **Workflow Scalability:** The combination of local QGIS analysis and GEE cloud-scale processing represents a robust professional standard for geospatial monitoring.

The final report and supporting documentation have been prepared for direct upload to the LMS portal as per the India Space Academy submission guidelines.

## 9 References

1. **USGS Earth Explorer:** Access point for Landsat-8 and Sentinel-2 MSI data (<https://earthexplorer.usgs.gov/>).
2. **Copernicus Open Access Hub:** Source for high-resolution multispectral reflectance datasets.
3. **GEE Documentation:** Technical guides for the smileRandomForest classifier (<https://developers.google.com/earth-engine/>).
4. **GADM Database:** Source for administrative boundary shapefiles for India (<https://gadm.org>).
5. **Breiman, L. (2001):** "Random Forests," Machine Learning, 45(1), 5–32.
6. **India Space Academy:** Instructional modules and project guidelines, Winter Training Program 2026.



# Rajalakshmi Engineering College Thandalam, Chennai.

## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

### Internship Feedback Form

#### Student Details:

Registration Number : 21 1 6 - 2 4 1 8 0 1 2 1 4

Student Name : Pr a v e e n R a m a n a n K

Department : Artificial Intelligence and Data Science

Year / Semester : I I / I V \_\_\_\_\_

Internship Organization : Indian Space Academy

Trainer Name : A l i s h a S i n h a

Internship Period : FROM : 3/01/2026 \_\_\_\_\_ TO: 28/02/2026 \_\_\_\_\_

Internship Duration : \_\_\_\_\_ 42 \_\_\_\_\_ DAYS

#### Section A: Student Feedback (to College)

1. Overall, how would you rate your internship experience?

☒ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

2. Did the internship help you gain useful knowledge or skills?

☒ Yes ☐ Partially ☐ No

3. What did you like most about your internship?

The most interesting aspect of my internship was working with real-time satellite data and using GIS tools like QGIS and Google Earth Engine. I particularly enjoyed applying the Random Forest algorithm to classify land use patterns, which gave me practical exposure to how machine learning is used in environmental analysis.

4. Remarks:

GOOD \_\_\_\_\_  
\_\_\_\_\_

Student Signature: Praveen Ramanan K \_\_\_\_\_ Date: 28/02/2026 \_\_\_\_\_

## Section B: Organization Feedback (to College)

1. Student's overall performance:

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

2. Attitude, discipline, and professionalism:

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

3. Contribution to assigned tasks/projects:

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

4. Comments / Recommendations:

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Organization Mentor/Manager Name & Designation: Alisha Sinha\_\_\_\_\_

Signature & Date: Alisha 28/02/2026\_\_\_\_\_



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अंतरिक्ष शिक्षा विभाग, भारत अंतरिक्ष सप्ताह  
Department of Space Education, India Space Week



Certificate No.:ISA/WT/2026/1008800

Enrolment No.: 042650

## Certificate

This is to certify that **PRAVEEN RAMANAN K** has been awarded this certificate on having completed the "Winter Training Programme on Remote Sensing and GIS" The training was conducted by **India Space Academy (ISA)** during 5th January 2026 to 21st January 2026. The participant has attended the training from **RAJALAKSHMI ENGINEERING COLLEGE**. During the training programme, the participant demonstrated good behaviour, discipline, and maintained proper decorum throughout the sessions.  
We appreciate the participant's dedication and extend our best wishes for their future academic and professional endeavours.

  
Program Supervisor  
India Space Academy

  
Affiliation Regulatory Authority  
India Space Week

  
Director  
India Space Academy

